LinUCB Is (Almost) All You Need: A Regime Analysis of LLM Warm-Starts in Contextual Bandits

(/pdf?id=0o1IySRk3t)

Muyi Su Lin (/profile?id=~Muyi_Su_Lin1), Solomon Xu (/profile?id=~Solomon_Xu1), Marissa Zhao Li (/profile?id=~Marissa_Zhao_Li1), Helen Yuan (/profile?id=~Helen_Yuan1), Xutong Liu (/profile?id=~Xutong_Liu1), William Chang (/profile?id=~William_Chang2)

24 Apr 2026 (modified: 27 May 2026) NeurIPS 2026 Conference Submission
Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors
Revisions (/revisions?id=0o1IySRk3t) CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/)

Abstract:
A recent line of work reports striking `reward multiplication'': combining a frozen LLM policy with an adaptive contextual bandit dominates either component, sometimes by an order of magnitude. We argue this is a \emph{regime indicator}, not a generic phenomenon---` cold-start LinUCB is (almost) all you need.'' We isolate three structural axes---effective dimension versus horizon, \textbf{feature coherence}, and \textbf{action-gap structure}---and prove a ``LinUCB-suffices'' bound along each. Saturating axis~~A or~~B alone certifies that no warm-start can improve cold-start LinUCB by more than a constant factor; axis~C provides a gap-dependent save when the effective dimension is moderate. The three-way separation behind the multiplication story lives where axes A and B jointly deviate; a regret-balancing analysis sharpens this to a phase transition at $r \asymp \sqrt{T}/\log T$. Four pre-runtime diagnostics classify any instance into one of four regimes before any algorithm runs; across every configuration we test, they correctly predict the order of magnitude of the warm-start lift.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.
Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.
Primary Area: General machine learning (e.g., core contributions in supervised and unsupervised methods), Decision making, reinforcement learning, and control (e.g., hierarchical RL, multi-agent systems, agentic AI)
Secondary Area: Deep learning advancements (e.g., architectures, optimizers, representation learning), Language and multimodal language models (e.g., text generation, summarization, VQA)
Contribution Type: Theory: The main contribution is via theoretical analyses and proofs.
Academic Integrity: I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (https://neurips.cc/Conferences/2026/MainTrackHandbook (https://neurips.cc/Conferences/2026/MainTrackHandbook)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.
LLM Usage: Prefer not to declare (you can then skip the rest)
LLM Experiment: Opt in to include this paper in the LLM-assisted peer review experiment.
Declaration: I confirm that the above information is accurate.
Reviewer Nomination: William Chang (/profile?id=~William_Chang2)
Ready For LLM Feedback: The submitted PDF is ready for LLM feedback.
LLM Feedback Request: Marissa Zhao Li (/profile?id=~Marissa_Zhao_Li1)
Submission Number: 4984

Citation Checker by Program Chairs

Citation Checker by Program Chairs 04 Aug 2026, 11:16 (modified: 04 Sept 2026, 22:11)

Program Chairs, Senior Area Chairs, Area Chairs, Authors Revisions (/revisions?id=ZvTyK3bNev)

Comment:

Automatically flagged references

Here we list the references that were automatically flagged by our checker. Note that our checker may miss many hallucinated references, and some of the listed references may have been flagged erroneously. **It is therefore crucial that you check them manually, following the guidance above. We rely on the Area Chairs to decide whether these constitute violations of responsible authorship.**

For papers falling into either of the following categories, we expect your default recommendation to be a rejection on the grounds of irresponsible authorship. This is strong presumption rather than a strict rule; as the Area Chair, you retain discretion, and may recommend otherwise where the specific circumstances of a paper warrant it:

1. The paper contains at least two hallucinated references according to the criteria described below the report.
2. The paper contains one hallucinated reference and there is some additional evidence of AI slop.

If a paper has only one problematic reference, but no additional concerns about the quality of the paper or its presentation, please simply alert the authors in your meta-review.

Please complete the form “Hallucinated References Investigation” to communicate the results of your investigation prior to completing your meta reviews.

Citation Checker Report

Reference Check Summary

0/60 problematic references.

Hallucinated/problematic reference criteria

These are adopted from ICLR and ICML 2026 hallucinated reference guidance:

- If you are unable to find any document or publication with a title anywhere near the one provided in the reference, it should be considered hallucinated. Minor differences in title should not be considered evidence of a hallucinated reference.
- If the reference's authors are only slightly incorrect, e.g., a few subtle misspellings, missing a few authors, or wrong authorship order, we do not consider this to imply a hallucinated reference, but please let the submission's authors know in your meta-review that they should fix this issue.
- If the author list has additional issues, e.g., added authors that are not actually authors of that publication or missing a large fraction of authors, then the reference should be considered hallucinated.

- If the reference's venue is a real venue but is incorrect for that reference, we do not consider this to imply a hallucinated reference, but please tell the submission's authors in your meta-review, these include wrong journal name, arXiv ID, etc.
- If there is no evidence that the reference's venue exists, the reference should be considered hallucinated.
- Finally, we note that there are many generic references that refer to existing artifacts (e.g. "GPT-4o, OpenAI et al. 2024") but do not refer to any real document. These are fine.
- Even with the above guidelines in mind, there will likely be some borderline cases. Please bring those up with your SAC if you need additional guidance.

Official Review of Submission4984 by Reviewer 6dJQ

Official Review by Reviewer 6dJQ 30 Jun 2026, 00:17 (modified: 04 Aug 2026, 06:07)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 6dJQ

Revisions (/revisions?id=fACldbaIoW)

Summary:
The paper addresses an important question: when do LLM-bandit hybrids improve over a strong adaptive baseline, rather than merely adding complexity? The paper proposes three structural axes: (A) effective dimension, (B) feature coherence, (C) action gap. The main thesis is that reward multiplication is not generic, but occurs only in specific regimes characterized by large effective dimension, low feature coherence, and weak/non-useful action-gap structure. The paper provides theoretical evidence supporting this main argument. The paper also validates this empirically in which most tested configurations show only marginal or negative warm-start lift, while one AmazonCat configuration with BERT features and Flan-T5-large reportedly achieves a $2.34\times$ lift over cold-start LinUCB.

Contribution Type: Theory: The main contribution is via theoretical analyses and proofs.

Strengths And Weaknesses:

Strengths.

- The paper studies when LLM warm-starts provide genuine multiplicative gains over cold-start LinUCB in linear contextual bandits. This is a well-posed question.
- The proposed axes (effective dimension, feature coherence, and action-gap structure) are intuitive and connect naturally to classical LinUCB/OFUL confidence geometry.
- The paper attempts to give both theoretical and empirical evidence to the central claim that the LLM warm-starts are only beneficial under certain specific scenarios.
- The paper validates empirically in one configuration with AmazonCat (BERT features) and Flan-T5-large reportedly achieves a $2.34\times$ lift over cold-start LinUCB. The negative results on WikiLinks and AmazonCat-BGE are valuable because they suggest that LLM warm-starting can be unnecessary or even harmful in many regimes.
- The paper suggests that the diagnostics are measurable offline with stated concentration guarantees. The pre-runtime diagnostic idea is practically appealing: deciding whether to deploy an expensive LLM-assisted router before running a long online trajectory would be useful if reliable.

Weaknesses:

- **Regret rate discrepancy.** The paper analyzes an OFUL/LinUCB-style algorithm whose confidence radius contains $\beta_T = \Theta(\sqrt{r \log T})$, yielding $\widetilde{O}(r\sqrt{T})$ regret, not $\widetilde{O}(\sqrt{rT})$. Since r is the central scaling variable, the extra $\sqrt{r}$ cannot be hidden in $\widetilde{O}(\cdot)$. The claimed $\widetilde{O}(\sqrt{rT})$ "LinUCB-suffices" rate would require a different algorithm or stronger assumptions.
- **Regret threshold error.** For the stated OFUL/LinUCB algorithm, the regret bound is $\widetilde{O}(r\sqrt{T})$, not $\widetilde{O}(\sqrt{rT})$. Counting both the confidence-radius and elliptical-potential logarithms gives a linear-regret threshold near $r \asymp \sqrt{T}/\log T$. Appendix B.4 only obtains a sublinear-looking $T/\sqrt{\log T}$ term by dropping the elliptical-potential $\sqrt{\log T}$ factor, so Lemma 1's stated rate and phase-transition interpretation are not justified.
- **Lemma 7 validity concerns.** Lemma 7 appears invalid as written. The cold-start lower bound invokes a minimax lower bound, which only guarantees the existence of a hard parameter instance, not an $\Omega(\sqrt{T})$ lower bound for the fixed constructed instance with $\theta^\star = e_1$. Moreover, the assumed isotropic warm start $V_0 = \lambda I + \nu\tau I$ is not shown to be realizable by low-regret expert-routed actions; in the proposed action set, obtaining coverage in the

e_2 direction requires pulling arms with non-negligible gap. Finally, the claimed polynomial shrinkage relies on replacing a logarithmic determinant-ratio improvement by $\sqrt{\lambda/(\lambda+\nu\tau)}$, which does not follow from the bound.

- **Insufficient regime characterization.** The three axes are presented as an exhaustive regime partition, but the results establish only one-directional sufficient conditions for cold-start LinUCB to have favorable upper bounds. The complement of these sufficient conditions is not proved sufficient for warm-start-essential behavior. Proposition 1/Lemma 6 show only that the joint-failure corner is nonempty. Moreover, the separate Q1–Q3 conditions and the ambiguous diagnostic bucket indicate that joint structural failure alone does not characterize when multiplication occurs.
- **Definitional imprecision.** The definition of "warm-starting is essential" via $R_T^{\mathrm{cold}}/R_T^{\mathrm{warm}} \to \infty$ should explicitly specify the asymptotic object: a sequence of instances indexed by T, possibly with growing d_T, r_T and warm starts (M_T, c_T) generated from $N_T = O(T)$ logged rollouts. This matters because the main separation relies on a growing-dimension construction, e.g. $d_T \geq T+1$, and the comparison is otherwise not well-defined.
- **Diagnostic classifier inconsistency.** The diagnostic classifier appears internally inconsistent. Appendix D.3 states that large subspace mismatch, e.g. angle above 45°, should output LinUCB-suffices because the warm-start does not cover the online/reward-relevant subspace. However, Table 1 reports an 85.1° mismatch for the main positive AmazonCat BERT/Flan-T5 configuration while classifying it as warm-start-essential. This contradicts the stated classifier and calls into question the claim that the diagnostics correctly predict the observed regimes.

Quality: 2: not good
Clarity: 2: not good
Significance: 3: good
Originality: 3: good
Questions:

- If the regret rate for the stated LinUCB algorithm is $\widetilde{O}(r\sqrt{T})$ rather than $\widetilde{O}(\sqrt{rT})$, which parts of the theory remain valid without modification? Please identify which lemmas/theorems require revision and provide the corrected dimensional thresholds.
- The current proof of Lemma 7 seems to require a substantially different argument: a fixed-instance lower bound for cold-start LinUCB plus a realizable low-regret warm-start that provides the claimed covariance coverage. Can the authors clarify whether such a construction exists, and whether the stated theorem/decision tree would change if Lemma 7 is removed or weakened?

Limitations:
The paper acknowledges several limitations, including the linear-reward assumption, fixed reward-relevant subspace, dependence on held-out data for the principal-angle diagnostic, and limited multi-expert evaluation. However, the limitations section should also acknowledge the gap between the theory and the diagnostic claims: the three-axis results provide sufficient conditions and selected separations, not an exhaustive characterization of when warm-starting is essential. Thus the empirical claim that the diagnostics classify all tested instances should be presented as preliminary evidence rather than as a theoretically certified classifier.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
No formatting concerns

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Rebuttal by Authors

Rebuttal
by Authors (Solomon Xu (/profile?id=~Solomon_Xu1), Helen Yuan (/profile?id=~Helen_Yuan1), William Chang (/profile?id=~William_Chang2), Marissa Zhao Li (/profile?id=~Marissa_Zhao_Li1), +2 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission4984/Authors))
25 Jul 2026, 21:44 (modified: 28 Jul 2026, 05:53)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=FcoS62zUll)

Rebuttal:
We thank the reviewer for the careful and detailed reading of our paper. The weaknesses and questions are addressed below.

Regret rate discrepancy ($\widetilde{O}(r\sqrt{T})$ vs. $\widetilde{O}(\sqrt{rT})$). We appreciate the reviewer raising this point. The notation $\widetilde{O}(\sqrt{rT})$ used in the lemma statements follows the convention in the contextual bandits literature (e.g., Abbasi-Yadkori et al. 2011, Lattimore & Szepesvári 2020) where $\widetilde{O}$ absorbs not only polylogarithmic factors but also the confidence radius β_T, which itself contains a $\sqrt{r \log T}$ factor from the self-normalized bound. This is explicitly unpacked in the proof of Lemma 1 (Appendix B.4, lines 681–687), where we write the full bound $R_T^{\text{cold}} \le C\beta_T\sqrt{rT\log(1+L^2T/\lambda)}$ and substitute $\beta_T = O(\sigma\sqrt{r\log T} + \sqrt{\lambda}S)$, obtaining the explicit rate $O(\sigma r\sqrt{T}\log T)$. The phase transition $r \asymp \sqrt{T}/\log T$ in Theorem 2 is derived from this fully expanded expression (Appendix B.9, Step 3, lines 908–923), so it correctly accounts for the $\sqrt{r}$ factor inside β_T. We agree that using $\widetilde{O}(\sqrt{rT})$ without clarifying what the tilde absorbs can be confusing, and we will make the convention explicit in the revision by adding a remark after the first appearance that "$\widetilde{O}(\sqrt{rT})$ absorbs both β_T and polylogarithmic factors; the fully explicit bound is $O(\beta_T\sqrt{rT\log(1+L^2T/\lambda)})$." We emphasize that all proofs, the phase-transition derivation, and the diagnostic thresholds work with the fully expanded form, so no result changes.

Regret threshold error. This concern is a consequence of the notational issue above. The reviewer states the threshold should be $r \asymp T/\sqrt{\log T}$ rather than $r \asymp \sqrt{T}/\log T$. However, the derivation in Appendix B.9 (lines 908–918) substitutes $\beta_T = O(\sqrt{r\log T})$ into $\beta_T\sqrt{rT\log T}$ and equates the result to $\Theta(T)$, yielding $r^{3/2}T^{1/2}(\log T)^{3/2} = \Theta(T)$, i.e., $r = \Theta(T^{1/3}/\log T)$... We acknowledge the algebra here needs careful re-examination due to how logarithmic factors interact. In Appendix B.4 (lines 682–685), the calculation proceeds: under $r\log T \le c\sqrt{T}$, we get $\beta_T = O(T^{1/4})$ and $\sqrt{rT} = O(T^{3/4}/\sqrt{\log T})$, so $\beta_T\sqrt{rT} = O(T/\sqrt{\log T})$, which is sublinear. Conversely, when $r = \omega(\sqrt{T}/\log T)$, $\beta_T\sqrt{rT} = \omega(T)$, so cold-start regret is linear and the bound is vacuous. The threshold $r \asymp \sqrt{T}/\log T$ comes from equating the exponents in this fully expanded expression. The appendix proof (B.4) does not "drop the elliptical-potential factor"—it uses the standard bound with $\gamma_T \le r\log(1+L^2T/(r\lambda))$ throughout. We will add an explicit step-by-step substitution in the revision to make this chain of inequalities fully transparent.

Lemma 7 validity concerns. We thank the reviewer for this careful criticism. We address the three sub-points:

(i) *Minimax lower bound on a fixed instance.* The reviewer correctly notes that a generic minimax lower bound guarantees the existence of a hard instance, not hardness of a specific one. Our construction is not generic: the 2D instance with $\theta^\star = e_1$ and action sets $e_1, u_t e_1 + \sqrt{1-u_t^2}, e_2$ where $u_t \sim \text{Unif}[0,1]$ is specifically designed so that a constant fraction of rounds have vanishingly small gaps (the gap is $1-u_t$, which is below η with probability η). The cold-start $\Omega(\sqrt{T})$ lower bound on this instance follows from information-theoretic arguments specific to 2D linear bandits with random action sets—not from a generic minimax invocation. We will replace the citation of the generic minimax theorem with a direct information-theoretic argument for this instance.

(ii) *Realizability of the isotropic warm-start.* The warm-start $V_0 = \lambda I + \nu\tau I$ can be realized by expert-routed actions that pull both arms in proportion. Because the second arm has a random feature direction that rotates over rounds, pulling it across τ rounds produces a design matrix whose expectation is $\tau \cdot \mathbb{E}[x_{t,A_t}x_{t,A_t}^\top]$, which is proportional to I_2 when the expert randomizes between the two arms. The gap on the second arm is $1-u_t$, which has mean $1/2$, so the expected instantaneous regret of the expert phase is $\epsilon_0 = 1/2$—positive but bounded. This satisfies Assumption A with $\epsilon_0 = O(1)$.

(iii) *Polynomial shrinkage claim.* The determinant ratio under the warm-start is $\log\det(V_{T+1})/\det(V_0) \le r\log(1+T/(\lambda+\nu\tau))$. Substituting $\nu\tau = \lambda T^\alpha$ gives a $\log(1+T^{1-\alpha}/\lambda)$ factor, which is $O(\log T)$—the same order as the cold-start factor $\log(1+T/\lambda)$. The shrinkage comes

from the $\sqrt{(T-\tau)/T}$ prefactor and the reduced effective regularization, not from a qualitative change in the logarithmic factor. We will add this clarification to the revised proof.

Insufficient regime characterization. We agree that our results provide sufficient conditions for LinUCB to be competitive, not a full equivalence. The "warm-start-essential" leaf is the complement of three sufficient conditions; we show this complement is nonempty (Proposition 1, Lemma 6) and that instances in it can exhibit diverging multiplicative gains, but we do not claim that every instance in the complement requires warm-starting. The Q1–Q3 conditions and the "ambiguous" bucket are intentionally part of the design: Q1–Q3 govern whether the deployed meta-algorithm *realizes* the gain (a separate question from whether the instance *permits* it), and the ambiguous bucket is a safety net for borderline cases. We will revise the language in Section 3.1 and the conclusion to state explicitly that the decision tree provides sufficient conditions for LinUCB competitiveness and sufficient conditions for warm-start gains at the joint corner, but does not claim exhaustive characterization of the boundary.

Definitional imprecision. The reviewer requests that $R_T^{\text{cold}}/R_T^{\text{warm}} \to \infty$ be defined over a sequence of instances indexed by T. This is correct, and it is how the proofs operate: Proposition 1 uses $d = n+1$ with $n = T$, and Lemma 6 uses $r = T$. We will add a paragraph to Section 2 defining the asymptotic comparison formally as a sequence (d_T, r_T, T) with (M_T, c_T) from $N_T = O(T)$ rollouts.

Diagnostic classifier inconsistency. The reviewer identifies a genuine presentation error: Appendix D.3 states that subspace mismatch above $\angle_0 = 45^\circ$ should trigger LinUCB-suffices, yet Table 1 reports $\sim 85^\circ$ mismatch for the AmazonCat BERT configuration classified as warm-start-essential. This is a mistake in how the deployment rule is stated. In the experiments, the subspace mismatch diagnostic is used as a *necessary* LLM-quality check (large mismatch warns that the LLM may not provide useful coverage), not as a sufficient condition for LinUCB-suffices. The actual classifier used in the experiments does not override the instance-level classification based on mismatch alone; instead, it flags high mismatch as a warning for Q2 (coverage). We will correct Appendix D.3 to match the implemented logic: the subspace mismatch enters the Q2 coverage check (Subsection D.5), not the instance-level deployment rule. We thank the reviewer for catching this inconsistency.

Questions

Q1: Which parts remain valid if the rate is $\widetilde{O}(r\sqrt{T})$? As discussed above, the proofs already operate with the fully expanded bound $O(\beta_T\sqrt{rT\log(\cdot)})$ where $\beta_T = O(\sqrt{r\log T})$. All lemmas and theorems are proved from this explicit form. The phase transition $r \asymp \sqrt{T}/\log T$ already accounts for the $\sqrt{r}$ inside β_T. The only revision needed is clarifying the $\widetilde{O}$ convention in the lemma *statements*—no proof, theorem, or diagnostic threshold changes.

Q2: Can the authors clarify whether a construction for Lemma 7 exists and whether the decision tree changes if it is removed? As discussed above, we believe the construction is valid but the lower-bound argument needs to be made instance-specific rather than citing the generic minimax theorem. If Lemma 7 were removed, the decision tree's axis C branch would lose its sharp-threshold companion, but the positive direction (Lemma 3: uniform gap implies LinUCB-suffices) remains intact. The tree structure does not change: axis C still serves as a "gap-dependent save," and the warm-start-essential leaf still requires all three axes to fail. What would change is that we could no longer formally certify that the warm-start-essential leaf is nonempty via axis C failure alone—but Proposition 1 and Lemma 6 already certify nonemptiness via joint failure of axes A and B with axis C vacuous ($\Delta_{\min} = 1$ but $r = T$).

Official Comment by Reviewer 6dJQ

Official Comment by Reviewer 6dJQ 03 Aug 2026, 01:28

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

I think the authors for the detailed discussion on the queries that I had raised in my review.

The rebuttal is strongest on two matters:

- It appropriately concedes that the three axes are not exhaustive.

- It agrees to formalize the T→∞ sequence.

But my central technical concerns about the dependence of $\sqrt{R}$ factor and its implications for the remainder of the paper remain to be validated after the rewrite. I would lean towards accepting that it might work out with minor changes, but I can't accept it without revisiting the paper post-rewrite. So I would like to keep my score.

Official Review of Submission4984 by Reviewer hk3d

Official Review by Reviewer hk3d 26 Jun 2026, 01:41 (modified: 11 Aug 2026, 01:27)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hk3d

Revisions (/revisions?id=bONmlsFtFY)

Summary:
This paper investigates when warm-starting linear contextual bandits with frozen LLM-derived policies truly yields "reward multiplication" over both components. The authors identify three structural axes—effective dimension versus horizon, feature coherence, and action-gap structure—and prove "LinUCB-suffices" lemmas along each axis, with sharp complementary thresholds in the appendix, implying that substantial multiplicative gains arise only in a narrow joint-corner regime. They also introduce a deterministic confidence-gated meta-algorithm with a regret shrinkage factor that formalizes how early expert routing can geometrically warm-start the covariance, and propose four pre-runtime diagnostics that predict the regime and observed lift across two datasets and multiple LLM/embedding pairings.

Contribution Type: Theory: The main contribution is via theoretical analyses and proofs.

Strengths And Weaknesses:

Strengths

- The regime-indicator framework that decomposes reward multiplication into three independent structural axes is conceptually clean; it refines prior intuition into precise sufficient and near-necessary conditions.
- The deterministic confidence-gated router provides a concrete mechanism and analysis yielding a multiplicative shrinkage factor $\Psi(\tau, \nu, T)$, clarifying the geometric warm-start effect beyond CORRAL-style masters.
- The phase transition $r \approx \sqrt{T}/\log T$ that emerges in the average-reward multiplication theorem connects asymptotics with a practical diagnostic and is a compelling characterization of when warm-start can be fundamentally helpful.
- The pre-runtime diagnostics are computed prior to any online interaction and used to predict regimes; the predictions match the sign and order of magnitude of observed gains across multiple configurations, including negative lifts.
- The study includes a parameter sweep for the router confidence on one dataset to rule out hyperparameter confounds in the "no-multiplication" cases.
- The AmazonCat-13K result demonstrates a substantial gain (2.34×) precisely where all three axes deviate, lending credence to the theory's joint-corner necessity.

Weaknesses

- Some assumptions in Theorem 1/2 are strong and hard to verify or enforce online (e.g., uniform early quality, existence of a reward-relevant subspace P with coverage parameter ν, and tail suboptimality of frozen experts), which may limit the directness of the guarantees.
- The coherence condition in Lemma 2 (uniform positive cosine similarity across all pairs of actions at each round) is very stringent; while it is presented as a sufficient condition, its worst-case nature may make the lemma less informative for many realistic high-K settings.
- Lack of head-to-head comparison with prior routing/aggregation approaches (e.g., CORRAL-based masters used in [13]) leaves open whether the deterministic router is competitive in practice.
- On AmazonCat with BGE-1024d features, the router collapses to 100% LLM routing at the chosen c; the claim that this is "a hyperparameter issue" would be strengthened by showing that an alternate c recovers LinUCB-dominant behavior, even if the regime says warm-start is unnecessary.

- Only two datasets are used (albeit with multiple configurations). Additional synthetic stress tests that isolate each axis independently (e.g., controlled coherence/gap sweeps) would better validate the sufficiency/necessity picture.
- The paper is motivated by LLM-based decision making, which seems to imply a specialized setting where contexts and actions are language objects or language-derived representations. However, the paper formalizes the problem only as a standard linear contextual bandit and does not clearly state how this language-structured setting enters the formulation.
- The choice of LinUCB among many existing linear bandit algorithms is not sufficiently justified. It would be helpful to clarify whether the analysis is specific to LinUCB or extends to other confidence-based linear bandit algorithms such as OFUL, Thompson sampling, or algorithms with sharper minimax or gap-dependent guarantees.
- LinUCB and warm-started LinUCB are central components of the paper, but their descriptions are rather brief. A more self-contained explanation of LinUCB and warm-start variants, especially how the initialization affects the design matrix and confidence radius, would improve readability.

Quality: 3: good
Clarity: 2: not good
Significance: 3: good
Originality: 3: good
Questions:

1. Can you comment on the tightness of Lemma 3? Are there known conditions under which the gap-dependent regret scales like $\tilde{O}(r \log T/\Delta_{min})$, and if so, how would that strengthen or shift axis C's role in the decision tree?
2. How can practitioners estimate or lower-bound the ν coverage parameter online for the deterministic router, and can routing be adapted to explicitly maximize ν subject to limited early regret?
3. In AmazonCat with BGE-1024d features, could you provide a c-sweep (or adaptive c) showing that the router can recover LinUCB-dominant behavior, corroborating that LinUCB-suffices is achievable by the router with appropriate tuning?
4. How sensitive are the diagnostic thresholds ($c_0 = 0.1$ for $\hat{r} \log T/\sqrt{T}$ and $\alpha_0 = 0.1$ for coherence) across datasets? Did you perform sensitivity analyses showing robustness of the regime classification to threshold choices?
5. Could you empirically compare the deterministic router against (i) CORRAL-based routing and (ii) a naive warm-start that initializes LinUCB with offline LLM rollouts, to quantify the practical benefits/risks of your approach?
6. For the coherence axis, have you explored weaker average-case or localized coherence metrics (e.g., per-context top-k neighbors) that might provide more realistic sufficient conditions and closer alignment with your median-based diagnostic?

Limitations:
The authors do discuss some limitations, including the linear-reward assumption, the fixed reward-relevant subspace assumption, the need for held-out data to compute certain diagnostics, and the limited stress-testing of multi-expert behavior. However, the discussion could be strengthened by more explicitly addressing the scope of the language-based setting, the dependence on the choice of representation/embedding space, and whether the conclusions are specific to LinUCB or extend to other linear bandit algorithms. It would also be helpful to discuss more concretely the potential risks of deploying LLM-guided bandit systems in recommendation or ranking applications, especially with respect to inherited biases from pretrained models and logged data.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
I did not find any violation of formatting, anonymity, etc.

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Rebuttal by Authors

Rebuttal

by Authors (Solomon Xu (/profile?id=~Solomon_Xu1), Helen Yuan (/profile?id=~Helen_Yuan1), William Chang (/profile?id=~William_Chang2), Marissa Zhao Li (/profile?id=~Marissa_Zhao_Li1), +2 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission4984/Authors))

25 Jul 2026, 21:44 (modified: 28 Jul 2026, 05:53)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=aV5O92owbE)

Rebuttal:
We thank the reviewer for the thoughtful evaluation and constructive suggestions. The weaknesses and questions are addressed below.

Strong assumptions in Theorem 1/2. We acknowledge that Assumptions A (uniform early quality), B (reward-relevant coverage), and D (tail suboptimality) are strong. This is by design: Theorem 1 characterizes the *best-case* warm-start gain, and the assumptions are the conditions under which that gain materializes. The pre-runtime diagnostics (Section 3.2) and LLM-quality proxies (Appendix D.5) are specifically designed to give practitioners measurable surrogates for these assumptions before committing to deployment. In practice, one checks Q1 via held-out LLM accuracy, Q2 via subspace mismatch, and Q3 by verifying the LLM is not near-optimal on the full horizon. We agree these are not verifiable online in closed form, but this is inherent to the problem: the deployment decision must be made before the online trajectory begins.

Coherence condition stringency. We agree that the worst-case coherence condition $\langle x_{t,a}, x_{t,a'} \rangle \geq \bar{\alpha} |x_{t,a}||x_{t,a'}|$ for all pairs is stringent, and we note this explicitly in the paper (lines 123–125). The condition is a sufficient condition for the $\widetilde{O}(\sqrt{rT})$ rate; it is not necessary. In practice, we use the median-of-minimum-cosines proxy $\hat{\bar{\alpha}}$ (defined in Appendix D.1), which is a much weaker statistic. The gap between the theoretical sufficient condition and the empirical proxy is discussed in Appendix D (lines 125–126). Weaker average-case or localized conditions (e.g., average coherence over top-k neighbors) would likely yield a tighter theory—this is a natural extension we plan to pursue.

Lack of head-to-head comparison with CORRAL. This is a fair request. The CORRAL-style master of [13] uses a randomized Online Mirror Descent update for routing, while our router is deterministic. The key theoretical distinction is that our deterministic routing transfers geometric information (covariance updates) from expert-routed rounds to LinUCB's design matrix on every round, which drives the shrinkage factor Ψ. CORRAL's randomized routing provides a different guarantee (competing with the best base algorithm). An empirical comparison would isolate whether the observed gains come from the deterministic covariance transfer or from general meta-algorithm properties. We will include this comparison in the revision.

BGE-1024d router collapse. The reviewer asks whether an alternate c recovers LinUCB-dominant behavior. At $c = 100$ with unit-norm BGE features, the LLM uncertainty $u^{(1)} = 1/\sqrt{c} = 0.1$ is permanently below LinUCB's uncertainty $u^{(0)} \geq 1/\sqrt{\lambda} = 3.16$, so the Router always defers to the LLM. At smaller c (e.g., $c = 1$), $u^{(1)} = 1$, which is closer to $u^{(0)}$ early on, and LinUCB would take over sooner as V_t grows. We did not include a c-sweep for AmazonCat BGE in the paper, but the prediction is clear: smaller c should recover LinUCB-dominant behavior, matching the LinUCB-suffices classification. We will add this sweep in the revision.

Only two datasets. We agree that additional datasets (especially synthetic stress tests isolating each axis independently) would strengthen the empirical validation. The two-dataset design was intentional—AmazonCat and WikiLinks together provide eight configurations spanning both sides of the regime boundary—but controlled synthetic experiments (e.g., sweeping coherence $\bar{\alpha}$ from 0 to 1 with fixed r and T) would directly test the sufficiency of each axis. We will add synthetic experiments in the revision.

Language-structured setting not formalized. The reviewer correctly notes that the theory operates entirely in the standard linear contextual bandit framework. The "language" aspect enters only through how features are constructed: context and action embeddings from pretrained models are combined into joint features $\phi(x, a) \in \mathbb{R}^d$ via element-wise product. The theory applies to any feature construction, not

just language-derived ones. We chose to motivate the paper through LLM-based decision making because (i) the reward-multiplication phenomenon was first documented in that setting [13], and (ii) the structural axes (coherence, effective dimension) have natural interpretations for pretrained embeddings. We will clarify this in the introduction: the formal setting is a standard linear contextual bandit; the LLM enters only through the feature map and the frozen policy.

Why LinUCB specifically. The analysis is not specific to LinUCB. Any confidence-based linear bandit algorithm that maintains a design matrix V_t and uses confidence widths of the form $|x|_{V_t^{-1}}$ (including OFUL [1] and SupLinUCB [15]) admits the same elliptical-potential argument and the same warm-start shrinkage mechanism. Thompson sampling with a Gaussian prior would have an analogous covariance-contraction effect. We focus on LinUCB because it is the most widely used algorithm in the applied contextual-bandit literature and because the confidence-gated routing rule has a clean geometric interpretation in terms of V_t^{-1}-norms. We will add a remark noting this generality.

LinUCB and warm-start descriptions too brief. We will expand the description in Section 2, particularly the explanation of how the initialization $(V_1, b_1) = (\lambda I + M, c)$ affects the confidence ellipsoid geometry and why this translates into smaller exploration bonuses early on.

Questions

Q1 (Tightness of Lemma 3). The gap-dependent bound $\widetilde{O}(r^2 \log^2 T/\Delta_{\min})$ is standard for LinUCB-type algorithms. Tighter gap-dependent bounds of $\widetilde{O}(r \log T/\Delta_{\min})$ are achievable with more sophisticated algorithms (e.g., OFUL with improved analysis or gap-aware elimination). If such a bound held for LinUCB, axis C would become a stronger rescuer: the gap-dependent bound would be non-vacuous for larger r (up to $r = o(T/\log T)$ instead of $r = o(\sqrt{T/\log T})$), shrinking the warm-start-essential region. The decision tree structure would not change, but the boundary of the gap-suffices leaf would shift.

Q2 (Estimating ν online). The coverage rate ν is not directly observable online because it depends on the unknown projector P. However, the minimum eigenvalue of the empirical Gram $\sum_{t \in \mathcal{T}} {\text{exp}}} x_{t,A_t} x_{t,A_t}^\top restricted to the top-r eigenvectors of V_t is a computable proxy that converges to \nu \tau as \tau grows. Adaptive routing that explicitly maximizes \nu$ would require solving an optimal design problem at each round (choosing expert actions to maximize the minimum eigenvalue of the projected Gram), which is an interesting direction but computationally expensive for large action spaces.

Q3 (BGE c-sweep). As discussed above, we will add this in the revision. The prediction is that smaller c values will recover LinUCB-dominant behavior, confirming the LinUCB-suffices classification.

Q4 (Sensitivity of diagnostic thresholds). We used $c_0 = 0.1$ and $\alpha_0 = 0.1$ throughout without tuning. On our eight configurations, the regime classification is robust to threshold perturbation: the AmazonCat BERT configuration has $\hat{\tilde{\alpha}} = 0.04$ (well below any reasonable $\alpha_0 \in [0.05, 0.2]$) and $\hat{r} \log T/\sqrt{T} = 0.16$ (above $c_0 \in [0.05, 0.15]$), while all WikiLinks configurations have $\hat{\tilde{\alpha}} \geq 0.19$ and $\hat{r} \log T/\sqrt{T} \leq 0.05$. The gap between the warm-start-essential and LinUCB-suffices clusters is large relative to threshold variation. That said, a formal sensitivity analysis across a wider range of datasets would strengthen this claim, and we will include one.

Q5 (Empirical comparison with CORRAL and naive warm-start). We will add both baselines. The naive warm-start (initializing LinUCB with offline LLM rollouts) is a special case of our framework with no online routing phase (τ rollouts collected offline). The comparison with CORRAL will isolate whether deterministic routing provides practical advantages beyond the theoretical distinction.

Q6 (Weaker coherence metrics). We have explored average-case coherence (mean over pairs rather than minimum) informally. On the AmazonCat datasets, the mean coherence is substantially higher than the minimum (~ 0.3 vs. 0.04 for BERT), which would misclassify the warm-start-essential configuration. The median-of-minimum proxy we use is a compromise: robust to outlier actions (via median) while capturing the worst-case alignment relevant to the theory (via minimum). A localized version (top-k neighbors within each context) is an interesting relaxation—it would capture the fact that LinUCB only needs to distinguish the optimal arm from its nearest competitors, not from all K arms.

Final Assessment

Official Comment by Reviewer hk3d 03 Aug 2026, 10:34

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Thank you for the detailed response. The clarifications on the scope of the three-axis framework and the proposed revisions would improve the manuscript. However, my central concerns remain unresolved. In particular, matching upper bounds for cold- and warm-started LinUCB do not by themselves establish the claimed constant-factor comparison, and the asymptotic form $\Psi \asymp \sqrt{\lambda/(\lambda+\nu\tau)}$ does not generally follow from the logarithmic ratio in Eq. (4). These issues affect the core "LinUCB-suffices" and separation claims. Moreover, the requested BGE c-sweep, routing baselines, and synthetic stress tests remain planned additions rather than evidence available in the current submission.

As resolving these points requires substantial theoretical revision and additional experiments, I maintain my score.

Official Review of Submission4984 by Reviewer F8Be

Official Review by Reviewer F8Be 24 Jun 2026, 22:15 (modified: 10 Aug 2026, 22:48)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer F8Be

Revisions (/revisions?id=fAPIYsMnFR)

Summary:

The paper investigates the conditions under which warm-starting contextual bandit algorithms with LLM-generated information can outperform a cold-start LinUCB baseline. The authors introduce three structural conditions—effective dimension, feature coherence, and action-gap structure, that characterize settings where warm starts are beneficial and settings where LinUCB is already sufficient. They substantiate this regime-based perspective through theoretical regret bounds and experiments on real-world contextual bandit datasets.

Contribution Type: Theory: The main contribution is via theoretical analyses and proofs.

Strengths And Weaknesses:

The paper addresses an interesting and timely question: under what conditions can LLM-based warm starts improve contextual bandit learning relative to cold-start LinUCB? This is a practically important setting, and a rigorous theoretical treatment of this phenomenon would be valuable. However, in its current form, the contribution is hard to evaluate, as the writing is often unclear and there seem to be several inconsistencies or gaps in the theoretical analysis. I describe my main questions and limitations below.

Quality: 1: poor

Clarity: 2: not good

Significance: 1: poor

Originality: 3: good

Questions:

(1) Is the "effective dimension of the reward-relevant subspace" in Lemma 1 formally defined?

(2) Does r denote the exact rank of a subspace containing $\theta^\star$ and all $x_{t,a}$, an empirical covariance effective rank such as

$$\hat{r} = \frac{\mathrm{tr}(\hat{\Sigma})}{|\hat{\Sigma}|_{\mathrm{op}}},$$

or an information-gain/effective-dimension quantity?

(3) Could the paper define P and r explicitly in the preliminaries and distinguish the theoretical r used in Lemma 1/Theorem 1 from the empirical diagnostic $\hat{r}$ in Appendix D?

(4) Are the proofs of Lemma 10 and Lemma 11 missing? I may have missed them, but I could not easily locate where these lemmas are proved.

(5) Do the “LinUCB-suffices” lemmas establish lower bounds showing that no admissible warm start can asymptotically beat cold LinUCB, or do they only provide matching upper bounds? In particular, showing that both cold and warm LinUCB have the same order upper bound does not rule out the possibility that the warm-started algorithm has much smaller regret.

Suggestions:

1. The introduction raises an interesting conceptual direction, but it is difficult to follow because several key quantities are invoked before they are defined. In particular, terms such as (r), effective versus ambient dimension, feature coherence (\bar\alpha), the gap parameter, and the shrinkage factor $\Psi(\tau, \nu, T)$ are presented as though they are already standard. Since these quantities are central to the paper’s regime classification, it would be helpful to introduce them informally in the introduction and relate them to familiar LinUCB regret rates. In its current form, the reader has to wait until Sections 2–3 to fully interpret the main claims.
2. The paper should define r formally in the preliminaries and clearly distinguish whether it denotes an exact reward-relevant subspace dimension, an empirical effective rank, or the effective dimension appearing in LinUCB-style regret bounds.
3. Because the paper’s main contribution concerns LLM-based warm starts, its regret bounds should be parameterized by measurable warm-start quality—such as prior accuracy, subspace coverage, and robustness to bad hints—and should clarify whether the guarantees interpolate between improved rates for good warm starts and standard LinUCB rates for poor ones, as in the online-learning-with-hints and predictable-sequences literature; see, for example, Cutkosky et al. (2022), *Leveraging initial hints for free in stochastic linear bandits*, and Rakhlin and Sridharan (2013), *Online learning with predictable sequences*.

Limitations:
Yes.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
N/A

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Rebuttal by Authors

Rebuttal
by Authors (Solomon Xu (/profile?id=~Solomon_Xu1), Helen Yuan (/profile?id=~Helen_Yuan1), William Chang (/profile?id=~William_Chang2), Marissa Zhao Li (/profile?id=~Marissa_Zhao_Li1), +2 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission4984/Authors))
25 Jul 2026, 21:44 (modified: 28 Jul 2026, 05:53)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors
Revisions (/revisions?id=2f2SGKhQma)

Rebuttal:
We thank the reviewer for engaging with the paper and for the specific suggestions. The questions and suggestions are addressed below.

Q1: Is the "effective dimension of the reward-relevant subspace" formally defined? Yes. The effective dimension r is defined implicitly in Section 2 (line 57–58) as the dimension of the subspace $\mathrm{Im}(P)$ containing $\theta^\star$ and all feature vectors $x_{t,a}$, where P is the rank-r orthogonal projector introduced in Assumption B (Section 3.3, line 174). We agree this definition is introduced too late and too implicitly. In the

revision, we will add an explicit definition in the Preliminaries (Section 2): "Let $P \in \mathbb{R}^{d\times d}$ be the orthogonal projector onto the smallest subspace containing $\theta^\star$ and all feature vectors $x_{t,a}$. The effective dimension is $r := \mathrm{rank}(P)$."

Q2: What does r denote? In the theoretical results (Lemmas 1–3, Theorems 1–2), r denotes the exact rank of the projector P as defined above—the dimension of the smallest subspace containing $\theta^\star$ and all features. This is a structural property of the instance, not an empirical estimate. The connection to the effective rank $\hat{r} = \mathrm{tr}(\hat{\Sigma})/|\hat{\Sigma}|_{\mathrm{op}}$ (the diagnostic) is that $\hat{r}$ is a continuous relaxation of r computable from data: when the feature covariance has r large eigenvalues and $d - r$ negligible ones, $\hat{r} \approx r$. Lemma 10 (Appendix D.2) provides the concentration guarantee.

Q3: Define r and $\hat{r}$ explicitly in the preliminaries. We agree and will do this. The revision will define r (theoretical, exact rank of P) in Section 2 and $\hat{r}$ (empirical effective rank of $\hat{\Sigma}$) in Section 3.2, with an explicit remark distinguishing the two and noting that $\hat{r}$ concentrates around r under the conditions of Lemma 10.

Q4: Are the proofs of Lemma 10 and Lemma 11 missing? Lemma 10 and Lemma 11 are stated as "consistency lemmas" in Appendix D.2 with proof sketches rather than full proofs. Lemma 10's sketch (lines 1035–1040) outlines the argument via matrix Bernstein for $|\hat{\Sigma} - \Sigma|_{\mathrm{op}}$ combined with scalar Hoeffding for $|\mathrm{tr}(\hat{\Sigma}) - \mathrm{tr}(\Sigma)|$. Lemma 11 is a direct application of the Dvoretzky–Kiefer–Wolfowitz inequality (stated in lines 1043–1046). Both are standard concentration results applied to our setting. We will expand these into complete proofs in the revision.

Q5: Do the "LinUCB-suffices" lemmas establish lower bounds showing no warm start can beat cold LinUCB, or only matching upper bounds? This is an important clarification. The "LinUCB-suffices" lemmas (Lemmas 1–3) show that *both* cold-start and warm-start LinUCB achieve the same asymptotic upper bound, which means no warm-start can improve the *order* of the rate. For example, Lemma 1 shows both achieve $\widetilde{O}(\sqrt{rT})$, so the ratio $R_T^{\mathrm{warm}}/R_T^{\mathrm{cold}}$ is $O(1)$. This does not rule out constant-factor improvements—warm-starting can still shave a constant factor within the same asymptotic rate. We state this explicitly in lines 112–113: "In cumulative regret, warm-starting can still shave a constant factor." The complementary direction—showing that warm-starting *can* produce a diverging ratio in the complement—is established by the negative-result lemmas (Lemmas 5, 6, 7 in the appendix). Together, the positive and negative results delimit where the ratio is $O(1)$ versus $\omega(1)$.

Suggestions

Suggestion 1 (Introduction readability). We agree that key quantities (r, $\bar{\alpha}$, Ψ) are invoked before definition. We will add a brief informal paragraph after the contributions list that introduces each quantity with a one-sentence description and forward-references the formal definition.

Suggestion 2 (Formal definition of r). Addressed above in Q3.

Suggestion 3 (Connection to online-learning-with-hints literature). We thank the reviewer for pointing to Cutkosky et al. (2022) and Rakhlin & Sridharan (2013). These works study how predictable sequences or initial hints can improve regret bounds in online learning, which is conceptually related to our warm-start framework. The key difference is that our warm-start operates through the *design matrix* (geometric coverage) rather than through a prior on the parameter. In the predictable-sequences framework, the benefit comes from reduced effective loss variation; in our framework, it comes from reduced effective dimension of the confidence ellipsoid. We will add a discussion of this connection in the related work, noting that parameterizing our bounds by measurable warm-start quality (accuracy ϵ_0, coverage ν, and tail gap ϵ_{tail}) is analogous to parameterizing by hint quality in that literature.

Rebuttal Acknowledgment

Official Comment by Reviewer F8Be 01 Aug 2026, 13:38

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

I thank the authors for their responses.

I believe these revisions would improve the readability of the manuscript and facilitate a more thorough evaluation of the results. The missing proofs should also be included, as they are necessary for assessing the validity and significance of the work.

I encourage the authors to incorporate these changes into future versions of the manuscript.

LLM Feedback by Program Chairs

LLM Feedback by Program Chairs 07 May 2026, 06:34 Conference, Authors

Feedback:
Dear Author,

We hope you found the PAT automated pre-submission feedback helpful for your NeurIPS 2026 submission.

We are currently gathering feedback to gauge the quality and usefulness of the PAT Program. Could you spare a few minutes to share your experience in the link below? Your insights will directly help us improve the tool's accuracy and utility for future authors.

https://docs.google.com/forms/d/e/1FAIpQLSfkkPCth8reUIUNubfiEjyoNLBm3lweHItoYWFYyQAk344Fww/viewform?usp=pp_url&entry.1908739883=0o1IySRk3t&entry.1717434670=MID&entry.445444243=109&pageHistory=0,1 (https://docs.google.com/forms/d/e/1FAIpQLSfkkPCth8reUIUNubfiEjyoNLBm3lweHItoYWFYyQAk344Fww/viewform?usp=pp_url&entry.1908739883=0o1IySRk3t&entry.1717434670=MID&entry.445444243=109&pageHistory=0,1)

Thank you for your time, Paper Assistant Tool (PAT) Team

LLM Feedback by Program Chairs

LLM Feedback by Program Chairs 05 May 2026, 04:49 Conference, Authors

Feedback:
Hello!

You requested a review of your paper submitted to NeurIPS using the Google Paper Assistant Tool (PAT). The resulting AI Feedback can be found below. Note that this feedback is posted automatically, and is only visible to authors. Importantly, the feedback will **not** be used in the review process. Reviewers, area chairs, and program committee members will **not** have access to the PAT feedback.

Disclaimer: Please note that the models used by the PAT pipeline are not infallible; they may hallucinate and make mistakes. Authors should treat the generated feedback with the same critical eye they would apply to a human review.

Summary

This paper provides a rigorous theoretical framework to explain the "reward multiplication" phenomenon observed when combining frozen Large Language Models (LLMs) with adaptive contextual bandits. The authors argue that this multiplicative gain is not a generic property but a specific regime indicator. They identify three structural axes—effective dimension versus horizon, feature coherence, and action-gap structure—and prove that a standard cold-start LinUCB algorithm will asymptotically match a warm-started hybrid up to a constant factor unless all three conditions deviate simultaneously. The paper introduces a deterministic, confidence-gated meta-algorithm that enables passive covariance updates, establishes a sharp dimensional phase transition for average-reward multiplication, and proposes four pre-runtime dataset diagnostics to predict when an LLM warm-start is essential.

Strengths

- The paper tackles a highly relevant and timely problem in the intersection of LLMs and sequential decision-making, offering a much-needed theoretical foundation for the empirically observed reward multiplication phenomenon.

- The identification of the three structural axes (effective dimension, feature coherence, and action-gap structure) and the proof of a sharp dimensional phase transition at $r \asymp \sqrt{T/\log T}$ are highly novel and significant contributions to the bandit learning literature.
- The proposed deterministic confidence-gated meta-algorithm elegantly isolates the geometric mechanism of warm-starting, showing how passive covariance updates from expert actions shrink the adaptive learner's confidence ellipsoid.
- Translating abstract theoretical conditions into four practical, pre-runtime dataset diagnostics is a strong contribution, allowing practitioners to predict the necessity and magnitude of LLM warm-starts before incurring significant computational costs.

Weaknesses

- Mathematical derivations in Lemma 2 (Feature Coherence): There appear to be issues in the proof of Lemma 2. Specifically, the proof squares the inequality $\langle u, v \rangle \geq \bar{\alpha}|u||v|$ for $-1 < \bar{\alpha} < 0$, which is algebraically invalid if the inner product is zero. Additionally, the assumption that the V_t^{-1} inner product preserves the Euclidean cosine lower bound $\bar{\alpha}$ up to a constant may be problematic, as the condition number of V_t grows linearly with t. The authors might consider revising this proof to ensure the geometric contraction argument holds.
- Logical consistency regarding Axis C (Action-Gap Structure) and Corollary 1: The paper suggests that a diverging multiplicative regret reduction requires the simultaneous failure of all three structural conditions. However, in the 'classical worst case' (Proposition 1), the optimal and suboptimal arms have expected rewards of 1 and 0, respectively, yielding a uniform gap ($\Delta_{\min} = 1$). This means Axis C is saturated, yet the warm-start regret is $O(1)$ while cold-start is n. The authors are encouraged to clarify this apparent contradiction regarding the necessity of Axis C's failure.
- Discrepancy in response vector (b_t) updates: There is a disconnect between the theoretical analysis, Algorithm 1, and the textual description. The text (e.g., Section 4.1) describes 'passive covariance updates' where b_t is updated only on LinUCB-active rounds. Conversely, Algorithm 1 unconditionally updates b_t, and the proof of Lemma 9 relies on a self-normalized concentration bound that assumes b_t includes rewards from all contexts contributing to V_t. If passive updates are used, the confidence radius would likely need to account for uncompensated bias. Please consider aligning the algorithm, theory, and text.
- Constructs and proofs in Lemmas 6 and 7: The authors might want to revisit the constructions in Lemmas 6 and 7. In Lemma 6, the defined features have non-zero components in $\mathrm{Im}(P^{\perp})$, which seems to violate Assumption B of Theorem 1. In Lemma 7, the 1D instance used to establish an $\Omega(\sqrt{T})$ lower bound for cold-start LinUCB appears to result in $O(1)$ regret, as the UCB score for the optimal arm would remain strictly positive and dominant.
- Derivations in Theorem 2: In the 'optimal balance' step of Theorem 2's proof (Lines 856-859), setting the derivative of the strictly convex function to zero locates its global minimum rather than its maximum. Furthermore, bounding the expected average reward using a high-probability bound on the realized regret without integrating over the failure event may require adjustment.
- Experimental setup for AmazonCat-13K: For the AmazonCat-13K experiment, the frozen policy uses an undefined internal confidence proxy ($\lambda_{\text{scale}}(1 - c_{\text{LLM}})$) instead of the cI_d design matrix. This substitution makes it difficult to determine whether the observed multiplier is driven by the geometric mechanism or the heuristic proxy. Providing an ablation with the standard cI_d proxy, as well as reporting variance over random seeds for this key dataset, would strengthen the empirical claims. Additionally, the classification of AmazonCat-13K as 'warm-start-essential' despite having $\bar{\alpha} = 0.04 \geq -0.5$ seems to contradict the deployment rule.
- Lack of empirical comparison with randomized baselines: While the paper distinguishes its deterministic routing rule from randomized CORRAL-style masters by highlighting the transfer of geometric information, there is no empirical baseline comparing the proposed Router against CORRAL. Including this comparison would help confirm that the observed performance gains are driven by the deterministic passive updates rather than general meta-algorithm properties (Chen et al.: Efficient Sequential Decision Making with Large Language Models, 2024).

Potential Issues And Suggestions

[Introduction and Related Work] (Pages: 1-3, 29-31)

1. Potential Mistakes and Improvements

- **Mathematical Error in Axis B (Feature Coherence) and Lemma 2:** The paper claims that linear regret requires $\bar{\alpha} \to -1$ (Corollary 1) and defines the "LinUCB-suffices" regime as $\bar{\alpha} \geq -0.5$ (Line 187, Figure 1). However, the proof of Lemma 2 (Appendix A.3, Line 625) contains a critical algebraic error. In Case 2 ($-1 < \bar{\alpha} < 0$), the proof assumes that $\langle u, v \rangle \geq \bar{\alpha}|u||v|$ implies $\langle u, v \rangle^2 \geq \bar{\alpha}^2|u|^2|v|^2$. This implication is invalid because the inner product can be zero (e.g., $0 \geq -0.5$ is true, but $0^2 \geq (-0.5)^2$ is false). Consequently, the claim that "The linear-regret instance of Proposition 1 cannot exist on any feature distribution with $\bar{\alpha}$ bounded above -1" (Line 161) is incorrect. In Proposition 1, the features are orthogonal, meaning the pairwise inner products are exactly zero, yielding $\bar{\alpha} = 0$. Since $0 \geq -0.5$, Proposition 1 demonstrates that cold-start LinUCB can suffer linear regret even when Axis B is "saturated."
- **Logical Contradiction Regarding Axis C (Action-Gap Structure) and Corollary 1:** The paper claims that for a warm-start to provide a diverging multiplicative regret reduction over cold-start LinUCB, *all three* conditions must fail simultaneously, including Axis C ($\Delta_{\min} \to 0$) (Corollary 1, Lines 176–182). This contradicts the paper's own "classical worst case" construction in Proposition 1, which the authors explicitly state "lives at the corner where all three deviate" (Line 30). In Proposition 1, the optimal arm has an expected reward of 1 and the suboptimal arm has an expected reward of 0, yielding a constant, uniform gap of $\Delta_{\min} = 1$ on every round. Thus, $\Delta_{\min} \not\to 0$. Despite Axis C being "saturated", cold-start regret is n while warm-start regret is $O(1)$, demonstrating that a warm-start can still be essential. The contradiction stems from interpreting Lemma 3's upper bound ($O(r^2 \log^2 T / \Delta_{\min})$) as a guarantee of low regret. When $r \asymp T$ (as required by Axis A's failure), this bound becomes $O(T^2 \log^2 T)$, which is vacuous. Sharing a vacuous upper bound does not preclude the actual cold-start and warm-start regrets from differing by a divergent factor. Therefore, Axis C's failure is not a necessary condition.
- **Theoretical vs. Empirical Disconnect Regarding the Response Vector (b_t) Updates:** There is a direct contradiction between the theoretical analysis, the proposed algorithm, and the textual claims regarding how the response vector b_t is updated during expert-routed rounds.
  - In Section 3.2, the theoretical analysis of the meta-algorithm (Theorem 1) relies on Lemma 9. The proof of Lemma 9 (Appendix A.8, Line 787) invokes the standard self-normalized concentration bound ($|\hat{\theta}_t - \theta^\star|_{V_t} \le \beta_t$). $For this bound to hold, the estimator \hat{\theta}_t = V_t^{-1} b_t$ must be computed using the rewards for *all* contexts that contribute to the design matrix V_t.
  - Algorithm 1 reflects this: Line 22 specifies that both V_{t+1} and b_{t+1} are updated for all rounds, regardless of whether the expert or LinUCB selected the action.
  - However, the text in the Introduction (Line 88: "our experiments use the passive convention $c = 0$") and Section 4.1 (Line 284: "the response vector b_t updates only on LinUCB-active rounds, matching Theorem 1's setting") claims that b_t is *not* updated for expert-routed rounds.
  - If V_t receives covariance updates from expert-routed rounds but b_t does not receive the corresponding rewards, the estimator $\hat{\theta}_t$ suffers a severe bias towards zero in the directions of the expert actions, violating the core confidence bound. This discrepancy should be resolved by correcting the text to align with the algorithm and theory. If the experiments intentionally omitted updating b_t, this introduces a substantial bias that requires explicit justification.

2. Minor Corrections and Typos

- **Preliminaries Notation ($V^{(i)}$):** In Section 2 (Line 60), the design matrix for the frozen policy is denoted as $V^{(i)}$. However, the index i is not defined until Section 3.2, where m experts are introduced. Omitting the index in the formal preliminaries would improve clarity.
- **Index Convention for V_t:** There is a notational inconsistency between Section 2 and Appendix A.1. In Section 2 (Line 62), $V_t = \lambda I + \sum_{s<t} x_{s,A_s} x_{s,A_s}^\top$ denotes the matrix *before* it is updated with round t's data. In the proof of Proposition 1 (Line 529), this exact same summation is denoted as V_{t-1}. Standardizing the time index convention would be helpful.

[Methodology and Meta-Algorithm] (Pages: 3-6, 24-29)

1. Potential Mistakes and Improvements

- **Contradiction Regarding "Passive" Updates and Confidence Radius Validity:**
  - The paper claims to isolate the geometric effect by using "passive covariance updates," stating that the response vector b_t "updates only on LinUCB-active rounds" and that the experiments use $c = 0$ (e.g., Lines 88, 284).
  - However, if V_t is updated with expert features while omitting the corresponding rewards from b_t, the LinUCB parameter estimate $\widehat{\theta}_t = V_t^{-1} b_t$ incurs a severe uncompensated bias. The bias term scales with $|M\{\text{exp}\} \theta^\star|_{V_t^{-1}} \approx \sqrt{\nu\tau}|\theta^\star|_P$.
  - The standard Abbasi-Yadkori confidence bound invoked in the proof of Theorem 1 (Lemma 9, Line 788) does not account for this bias. To maintain validity under passive updates, the confidence radius β_T would need to grow by $\mathcal{O}(\sqrt{\tau})$. This would result in a post-handover regret bound of $\mathcal{O}(\sqrt{\tau(T-\tau)})$, destroying the claimed $\mathcal{O}(\sqrt{rT})$ upper bound and the multiplicative shrinkage effect.
  - Conversely, Algorithm 1 (Line 22) actually implements *active* updates ($b_{t+1} \leftarrow b_t + r_t x_{t,A_t}$ is executed on every round). The authors should resolve this discrepancy. If active updates are required for the theory to hold, the assertions regarding a purely "passive" geometric effect should be revised.
- **Algebraic Error in Lemma 2 (Coherent Features) for $\bar{\alpha} \le 0$:**
  - In Appendix A.3 (Line 625), the proof states that for $-1 < \bar{\alpha} < 0$, the inequality $\langle u, v \rangle \ge \bar{\alpha}|u||v|$ implies $\langle u, v \rangle^2 \ge \bar{\alpha}^2 |u|^2 |v|^2$ "since the squared cosine is symmetric in sign."
  - This algebraic step is mathematically incorrect. One cannot square an inequality $x \ge y$ if $x = 0$ and $y < 0$ (e.g., $0 \ge -0.5$ does not imply $0^2 \ge 0.25$). For example, perfectly orthogonal features yield $\langle u, v \rangle = 0$, satisfying the condition for any negative $\bar{\alpha}$, but failing the squared inequality.
  - This error invalidates the proof of Lemma 2 for $\bar{\alpha} \le 0$. This directly contradicts Proposition 1, which demonstrates linear regret for orthogonal features (where $\bar{\alpha} = 0 \ge -0.5$). Consequently, the diagnostic deployment rule (Line 994) incorrectly classifies instances with $\bar{\alpha} \ge -0.5$ into the "LinUCB-suffices" regime.
- **Invalid Application of Theorem 1 in Lemma 6 (Violation of Assumption B):**
  - Theorem 1 explicitly requires Assumption B, which mandates that all action features lie entirely within the reward-relevant subspace: $x_{t,a} \in \mathrm{Im}(P)$ (Line 226). This ensures the determinant ratio factorizes cleanly across the subspaces in the proof (Line 802).
  - The construction in Lemma 6 (Line 665) defines features that explicitly have non-zero components in the orthogonal complement $\mathrm{Im}(P^\perp)$.
  - Despite this violation of Assumption B, the proof of Lemma 6 (Line 684) invokes Theorem 1 to claim an $\mathcal{O}(\sqrt{rT})$ warm-start upper bound. This is mathematically invalid; the presence of $P^\perp$ components prevents the determinant ratio from factorizing, and LinUCB would be forced to explore the $d - r$ nuisance directions, resulting in $\Omega(\sqrt{dT})$ regret even with the warm start.
- **Mathematical Error in Lemma 13 (Subspace Mismatch):**
  - Lemma 13 (Line 1032) claims that the subspace mismatch diagnostic provides a Loewner lower bound on the coverage matrix using the *mean* principal angle $\bar{\angle}$: $\sum \mathbb{E}[x_{t,A_t} x_{t,A_t}^\top] \succeq (\cos^2 \bar{\angle}) \nu_0 \tau P$.
  - The minimum eigenvalue of the projection onto P is determined by the *maximum* principal angle between the subspaces, not the mean. If a single direction in P is orthogonal to the expert's subspace (maximum angle $\pi/2$), the restricted minimum eigenvalue drops to 0, rendering the bound $\succeq (\cos^2 \bar{\angle}) \dots P$ false. The lemma should be corrected to rely on the maximum principal angle.
- **Algorithm 1 Notation and Missing Definitions:**
  - Lines 23–24 of Algorithm 1 invoke confidence bounds $U_t^{(i)}$ and $L_t^{(0)}$ to trigger a handover, but their formulas are missing from the pseudocode block.
  - The descriptive text (Line 931) defines these bounds using a multiplier $\sqrt{\beta_t}$. This is dimensionally inconsistent with the standard LinUCB parameterization (e.g., Line 65 and Line 790), which uses β_t directly as the confidence radius without a square root.
  - Algorithm 1 accepts α_t as input (Line 5) but the text refers to β_t. The symbols should be consolidated for clarity and reproducibility.

2. Minor Corrections and Typos

- **Lemma 10 Proof Sketch (Line 977):** The suggested perturbation bound $|\mathrm{tr}(\widehat{\Sigma}) - \mathrm{tr}(\Sigma)| \le d|\widehat{\Sigma} - \Sigma|_{\mathrm{op}}$ introduces a false dependence on the ambient dimension d. To achieve the r-dependent bound claimed in the lemma, the trace error should be bounded directly via scalar concentration on $\frac{1}{n}\sum |x_i|^2 \le L^2$.
- **Data Requirements for Diagnostics (Line 183 vs. Line 958):** Section 3.1 states that the diagnostics are measurable from an "offline sample of contexts." However, the Gap Distribution diagnostic explicitly requires fitting an oracle model "using ground-truth rewards." The text should clarify that this specific diagnostic requires a fully labeled offline dataset, not just contexts.
- **Missing Expectation in Assumption D:** In Section 3.3 (Line 246), the definition of the uniform tail gap lacks an expectation operator over the contexts, making it a random variable rather than a deterministic bound. It is written correctly with the expectation operator in Appendix D (Line 1017).
- **Direct Sum Notation (Line 685):** The text states that "action-set features lie in $\mathrm{Im}(P) \cup \mathrm{Im}(P^\perp)$". Because the features are linear combinations of vectors from both subspaces, they lie in the direct sum $\mathrm{Im}(P) \oplus \mathrm{Im}(P^\perp)$, not the union.

[Theoretical Results and Proofs] (Pages: 3-7, 13-24)

1. Potential Mistakes and Improvements

- **Invalid Asymptotic Approximation of the Shrinkage Factor Ψ (Theorem 1, Section 3.2 & Appendix A.8)**
  - **Issue:** The paper claims that for $\nu\tau \gg \lambda$, the multiplicative shrinkage factor $\Psi(\tau, \nu, T)$ scales as $\sqrt{\lambda/(\lambda + \nu\tau)}$ and vanishes as $\nu\tau \to \infty$ (Lines 235, 819-823). This appears mathematically incorrect.
  - **Explanation:** Equation (5) defines Ψ with the term $\sqrt{\frac{\log(1+L^2(T-\tau)/(\lambda+\nu\tau))}{\log(1+L^2T/\lambda)}}$. For any typical sublinear warm-start phase (e.g., $\tau = o(T)$), the arguments inside both logarithms grow linearly with T. The ratio of logarithms $\frac{\log(c_1 T)}{\log(c_2 T)}$ evaluates asymptotically to 1, not c_1/c_2. For instance, if $\tau = T^\alpha$ ($\alpha < 1$), the ratio approaches $\sqrt{1-\alpha}$, which is bounded strictly away from zero.
  - **Impact:** Ψ behaves as $\Theta(1)$ rather than $1/\sqrt{\nu\tau}$. This means the meta-algorithm achieves only a constant-factor reduction over cold-start LinUCB, which invalidates the central claim of an $\omega(1)$ multiplicative gain and disrupts the phase transition derived in Theorem 2.
- **Violated Assumption and Flawed Construction in Lemma 6 (Appendix A.5)**
  - **Issue:** The proof of Lemma 6 invokes Theorem 1 to bound the meta-algorithm's regret (Line 689). However, the construction explicitly violates a required assumption of Theorem 1, and the algebraic comparison of arms is incorrect.
  - **Explanation:** Theorem 1 relies on Assumption B (Line 226), which requires all action features to lie entirely in the reward-relevant subspace $\mathrm{Im}(P)$. In the Lemma 6 construction, actions g_t and b_t explicitly have components $w_{f(t)} \in \mathrm{Im}(P^\perp)$ (Line 664), rendering Theorem 1 inapplicable. Furthermore, the proof claims that "both arms have identical $\mathrm{Im}(P^\perp)$ components" (Line 687). Based on the definitions, g_t has a $\frac{1}{\sqrt{2}} w_{f(t)}$ component, while b_t has a $\frac{L}{\sqrt{2}} w_{f(t)}$ component. Since $L > 1$, they are not identical.
  - **Impact:** Because the arms have unequal $P^\perp$ components, warm-started LinUCB will suffer the exact same $\Omega(\sqrt{dT})$ optimism trap in $P^\perp$ as cold-start LinUCB. The claimed $\tilde{O}(\sqrt{rT})$ upper bound for the meta-algorithm is thus invalid on this instance.
- **Invalid Lower Bound Argument in Lemma 7 (Appendix A.6)**
  - **Issue:** The proof attempts to establish an $\Omega(\sqrt{T})$ lower bound for cold-start LinUCB on a 1D instance where $\theta^\star = 1$ and action sets are $u_t, 1$ with $u_t < 1$ (Lines 710–718).
  - **Explanation:** In 1D, the LinUCB upper confidence bound factors as $\text{UCB}*t(a) = a(\hat{\theta}*{t-1} + \beta_t / \sqrt{V_{t-1}})$. $Since the term in parenthese is an upper confidence bound for \theta^\star = 1 > 0$, $it is strictly positive with high probability. Thus, the \text{UCB}* score is strictly increasing in a$. $Because 1 > u_t, LinUCB will evaluate 1 \cdot (\text{positive}) > u_t \cdot (\text{positive})$ $and deterministically select the optimal arm a=1$ at almost every round.
  - **Impact:** LinUCB achieves $O(1)$ regret on this specific instance. Applying a generic minimax lower bound to this trivially solvable problem is mathematically invalid and breaks the proof of the lemma.

- **Invalid Geometric Contraction Argument in Lemma 2 (Appendix A.3)**
  - **Issue:** The proof asserts that the Mahalanobis inner product preserves the Euclidean cosine lower bound $\bar{\alpha}$ "up to its conditioning, which can be absorbed into $\bar{\alpha}$ at the cost of a constant depending on λ, L" (Lines 622–624).
  - **Explanation:** The condition number of the empirical design matrix $V_t = \lambda I + \sum x_s x_s^\top$ is bounded by $(\lambda + tL^2)/\lambda$, which grows linearly with t and is not a constant. In a highly anisotropic space, the V_t^{-1} inner product between two vectors can approach zero or become negative even if their Euclidean inner product is strictly positive.
  - **Impact:** If the V_t^{-1} inner product is ≤ 0, the contraction factor $(1 - \bar{\alpha}^2 \rho_t)$ evaluates to ≥ 1, meaning no contraction occurs. This breaks the recursive geometric contraction argument and invalidates the derived $\tilde{O}(\sqrt{rT}/\bar{\alpha})$ regret bound.
- **Omitted β_T Factor and Asymptotic Collapse in Lemma 1 (Appendix A.4)**
  - **Issue:** In Lines 636–639, the proof evaluates the cold-start regret under the boundary condition $r \log T \leq c\sqrt{T}$, incorrectly concluding $R_T^{\text{cold}} = \tilde{O}(\sqrt{rT}) = \tilde{O}(T^{3/4})$.
  - **Explanation:** The standard LinUCB regret bound is $\tilde{O}(\beta_T \sqrt{rT})$. The proof calculates $\beta_T = O(T^{1/4})$, but then drops β_T from the final expression. Substituting properly yields $R_T^{\text{cold}} = \tilde{O}(\beta_T \sqrt{rT}) \approx \tilde{O}(T^{1/4} \cdot T^{3/4}) = \tilde{O}(T)$.
  - **Impact:** The correct evaluation yields linear regret ($\tilde{O}(T)$), which directly contradicts the lemma's claim that cold-start LinUCB achieves sublinear regret ("suffices in average reward") in this regime.
- **Calculus Error in Optimal Balance (Theorem 2, Appendix A.9)**
  - **Issue:** In the proof of Theorem 2, Step 2 (Lines 856–859) attempts to find the "optimal balance" $\tau^\star$ that maximizes the regret gap lower bound $\Phi(\tau, T)$ by setting $\partial_\tau \Phi = 0$.
  - **Explanation:** The function $\Phi(\tau, T)$ can be written in terms of $x = T - \tau$ as $\Phi(x) \approx Ax - B\sqrt{x \log(1 + Dx)}$. Because the term $-B\sqrt{x \log(\dots)}$ is strictly convex with respect to x, the sum is strictly convex. Setting the derivative of a strictly convex function to 0 locates its global *minimum*, not its maximum.
  - **Impact:** By solving this balance equation, the proof actively minimizes the lower bound, yielding a vacuous result. As noted correctly earlier in the proof, the true maximum on the interval must occur at the boundaries (e.g., $\tau = 0$).
- **Invalid Mixing of Expectations and High-Probability Bounds (Theorem 2)**
  - **Issue:** In Section 3.3 (Line 245), the average reward $\bar{r}_T^{\text{alg}}$ is defined as the expectation $\frac{1}{T}\sum \mathbb{E}[r_t^{\text{alg}}]$, making it a deterministic scalar. Theorem 2(i) (Line 253) states a bound on this quantity holds "with probability at least $1 - \delta$."
  - **Explanation:** A deterministic quantity cannot be bounded "with probability at least $1 - \delta$." Furthermore, in the proof (Lines 842–844), the expectation of the regret is taken, but then bounded directly by substituting $G(\tau, T)$, which is a high-probability bound on the *realized* regret from Lemma 9.
  - **Impact:** Bounding an expectation using a high-probability event requires integrating over the failure event, which typically introduces an additional $\delta \cdot \Theta(T)$ term. The current formulation is a mathematical type error.
- **Mismatch Between Lemma 3 Statement and Proof (Appendix A.7)**
  - **Issue:** The statement of Lemma 3 (Line 172) claims a gap-dependent regret bound of $\tilde{O}(r^2 \log^2 T / \Delta_{\min})$.
  - **Explanation:** The algebraic derivation in the proof (Lines 748–750) bounds the total number of suboptimal pulls N_T by a factor proportional to $1/\Delta_{\min}^2$, and multiplies this by the maximum possible gap $2LS$. This mathematically yields a final bound of $\tilde{O}(r^2 \log^2 T / \Delta_{\min}^2)$. In the final equality (Line 750), the denominator is inexplicably changed to $\Delta_{\min}$ to match the lemma statement.
  - **Impact:** To legitimately achieve the claimed $1/\Delta_{\min}$ rate, the proof must bound the sum of the actual instantaneous gaps $\sum \Delta_{A_t}$ directly, rather than multiplying a global bound on N_T by a constant maximum gap.

2. Minor Corrections and Typos

- **Lines 64, 736, and 790:** The LinUCB selection rule is written as $\arg\max_a \langle \hat{\theta}*t, x*{t,a} \rangle + \beta_t |x_{t,a}|_{V_t^{-1}}. The estimates must be from the previous step (\hat{\theta}_{t-1} and V_{t-1}^{-1}), as the current step's estimates cannot be computed before selecting the action A_t$.

- **Line 109:** The notation `0e` appears to be a LaTeX compilation typo for $\tilde{O}$.
- **Line 604:** The Taylor expansion for the square root perturbation $\sqrt{1/\lambda + \Delta} - \sqrt{1/\lambda}$ is misstated algebraically as $\sqrt{\Delta} \cdot \sqrt{\lambda}/2$. It should be approximately $\Delta\sqrt{\lambda}/2$.
- **Line 638:** The notation for the fourth root of T is written as `\sqrt4 T`, which renders as $\sqrt{4}T$ rather than $T^{1/4}$.
- **Line 664 Indexing:** The bad arm definition uses $u_{(t \bmod (r-1))+1}$. The indexing cycles through $1, \ldots, r-1$, but earlier the orthogonal basis for $\mathrm{Im}(P)$ was defined as $u_0, \ldots, u_{r-1}$ (Line 662), meaning u_0 is skipped in the bad arms.

[Experimental Evaluation] (Pages: 7-9)

1. Potential Mistakes and Improvements

- **Undefined Variables and Confounded Routing Mechanism (AmazonCat-13K):** In Section 4.4, the experimental setup for AmazonCat-13K states that the frozen policy "uses an internal confidence proxy $\lambda_{\text{scale}}(1 - c_{\text{LLM}})$ in place of the cI_d design matrix used on WikiLinks" (Lines 331–332). First, the variables λ_{scale} and c_{LLM} are undefined in the text or appendices, making this setup difficult to reproduce. Second, altering the routing mechanism for the only dataset that yields the "warm-start-essential" regime introduces a significant confounding variable. The theoretical framework (Theorem 1) relies on geometric coverage through the design matrix (Assumption B). By substituting the geometric cI_d proxy with an LLM's internal confidence score, it becomes unclear whether the observed $4.83\times$ multiplier is driven by the structural regime's geometry (as theorized) or simply by the non-geometric heuristic proxy. Providing an ablation that evaluates AmazonCat-13K with the standard cI_d proxy would properly isolate the effect of the dataset's structural regime.
- **Ambiguity in Joint Feature Construction and Effective Rank (AmazonCat-13K):** Table 2 reports an effective-dimension diagnostic of $\hat{r}/\sqrt{T} \approx 2.6$ for AmazonCat-13K. Given the horizon $T = 1,186,239$ (Table 1), this mathematically implies an effective rank of $\hat{r} \approx 2831$. However, Section 4.4 specifies that SVD was applied to reduce action embeddings to 50 dimensions. If the joint feature construction $\phi(x, a)$ were formed via a dimension-preserving element-wise product (as defined for WikiLinks in Section 4.2, $\phi(x, a) = x \odot a$), its dimensionality would be strictly bounded by the 50-dimensional action space, making an effective rank of 2831 impossible. Explicitly defining how the joint features are constructed for AmazonCat-13K (e.g., via a Kronecker product) is necessary to clarify how this effective rank is achieved and to ensure reproducibility.
- **Contradiction in Regime Classification Logic (Axis B):** There is a contradiction between the deployment rule (Appendix C.3), the theoretical bounds, and the classifications reported in Table 2. Appendix C.3 (Line 994) states that an instance should be classified as `LinUCB-suffices` if $\bar{\alpha} \geq -0.5$. Table 2 reports $\bar{\alpha} = 0.04$ for AmazonCat-13K, which mathematically satisfies this condition ($0.04 \geq -0.5$); however, it is classified as `warm-start-essential`. Lemma 2 provides a cold-start regret bound of $\tilde{O}(\sqrt{rT}/\bar{\alpha})$, which becomes vacuous as $\bar{\alpha} \to 0$ (the near-orthogonal regime). Thus, the condition $\bar{\alpha} \geq -0.5$ incorrectly includes the orthogonal failure mode. This also contradicts Figure 1, which requires positive coherence ($\bar{\alpha} \geq \alpha_0 > 0$). The deployment rule should be corrected to align with the theoretical bounds and empirical predictions.
- **Mathematical Impossibility regarding τ (Batches vs. Rounds):** In Section 4.4 (Line 336), the text states that $\nu\tau$ is empirically estimated using "$\tau \approx 25,000$ batches." According to Section 4.1 (Line 276), the data stream is processed in batches of 128. If τ represents 25,000 batches, this equates to $3,200,000$ rounds, which is impossible as it vastly exceeds the total dataset horizon $T = 1,186,239$ (Table 1). It is highly likely $\tau \approx 25,000$ refers to *rounds*, not batches, but this discrepancy should be corrected.
- **Missing Variance Reporting for the Primary Stress Test:** Section 4.1 (Line 286) indicates that the general experimental setup uses "Three random seeds," and Table 3 consistently reports the mean and standard deviation across these seeds for the WikiLinks datasets. In contrast, the results for the AmazonCat-13K experiment (Section 4.4) are reported only as point estimates (e.g., "~ 0.04" for Cold LinUCB, "~ 0.285" for the Router). Because AmazonCat-13K is the sole configuration demonstrating the "warm-start-essential" regime and the multi-fold gain, reporting variance over the random seeds would help establish the statistical significance and stability of this central empirical result.

2. Minor Corrections and Typos

- **Line 331:** The variable `cLLM` is formatted inconsistently in the text; standardizing to math-mode formatting (e.g., c_{LLM}) would improve readability.

- **Lines 335–336:** The variables representing the meta-algorithm and fixed policy average rewards are written as $\bar{r}^{\text{meta}}/\bar{r}^{\text{fix}}$, missing the subscript T used during their theoretical definition in Section 3.3 ($\bar{r}_T^{\text{meta}}/\bar{r}_T^{\text{fix}}$).

[Conclusion, References, and Checklist] (Pages: 9-13, 31-37)

1. Potential Mistakes and Improvements:

- **Incomplete Submission Checklist:** The NeurIPS Paper Checklist (Pages 31–37) has not been populated. All answers and justifications are currently left as the template `[TODO]` placeholders (e.g., lines 1163, 1164, 1177, 1178, etc.). The authors should complete this checklist to comply with submission requirements.
- **Incomplete Reference:** Reference ("Contextual bandits with llm-derived priors and adaptive calibration", lines 424–425) is missing necessary publication details, including the publication venue (or preprint server identifier) and the year of publication.

2. Minor Corrections and Typos:

- **Bibliography Capitalization and Formatting:** Several references suffer from standard BibTeX capitalization stripping, resulting in lowercase acronyms and proper nouns. These can be fixed by wrapping the affected terms in curly braces `{}` in the BibTeX file. Examples include:
  - Reference: "Gpt-4" should be "GPT-4"
  - Reference: "llm-generated" should be "LLM-generated"
  - Reference: "llm-derived" should be "LLM-derived"
  - Reference: "Qwen2. 5-coder" contains an extraneous space and should be corrected to "Qwen2.5-Coder"
  - Reference: "Llms" should be "LLMs"
  - Reference: "chinese" should be "Chinese"
  - Reference: "llm" should be "LLM"

LLM Feedback by Program Chairs

LLM Feedback by Program Chairs 05 May 2026, 02:58 Conference, Authors

Feedback:

Hello!

```
You requested feedback on your paper submitted to NeurIPS using the Google Paper Assistan
t Tool (PAT).
We received your paper and are currently processing it. We will post the feedback here on
ce it is ready.
If you have not received feedback within 24 hours, please contact us at paper-assistant@g
oogle.com.
```

About OpenReview (/about) Contact (/contact)
FAQ (https://docs.openreview.net/getting-started/frequently-asked-questions) Hosting a Venue (/group?id=OpenReview.net/Support)
Sponsors (/sponsors) Terms of Use (/legal/terms) / Privacy Policy (/legal/privacy)
All Venues (/venues) **Donate** (/donate)
OpenReview (/about) is a long-term project to advance science through improved peer review with legal nonprofit status. We gratefully acknowledge the support of the OpenReview Sponsors (/sponsors). © 2026 OpenReview